

**AI-Driven Optimization of Satellite Communication
Networks Using Big Data Platforms:
An Integrated Reference Architecture for Data-Driven Constellation Control**

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1. Introduction

Satellite communications are entering a period of rapid densification. Low-Earth-orbit (LEO) megaconstellations such as Starlink, OneWeb, and the upcoming Kuiper deployments are placing thousands of satellites into orbit, and the broader satellite-communications market is projected to grow by the end of the decade (Ochuba et al., 2024; Kodheli et al., 2021). The result is a network whose state changes second by second. Satellites cross overhead in minutes, beams steer in seconds, interference is non-stationary, and user demand is bursty and geographically skewed. Operational tools built around periodic reports and static link budgets cannot keep up with this tempo.

Each satellite and gateway also produces large volumes of telemetry, traffic counters, and payload measurements. Ochuba et al. (2024) and Liu (2020) describe volumes at the petabyte scale per constellation per year, with velocities that place this data in the streaming regime. The literature already contains every component needed to close a data-driven control loop over such a system: ingest, streaming analytics, machine learning, and on-board edge execution. Those components remain siloed across separate communities. The thesis of this paper is that integrating them into a layered reference architecture built on Apache Kafka, Apache Flink, and Apache Spark, paired with three families of machine-learning models and partial on-board execution, is the most useful near-term direction for AI-driven satellite-network optimization.

2. Literature Review

The relevant literature falls into four streams. The first is survey and strategy work. Kodheli et al. (2021) gives a broad treatment of the new space era, covering physical-layer, networking, and regulatory dimensions. Ochuba et al. (2024) narrows the lens to AI-driven optimization of satellite networks and flags big-data integration as a recurring open problem.

The second stream is big-data platform research. Vyas et al. (2022) benchmarks Apache Kafka and shows that partitioned topics and replication provide linear scalability under high producer rates. Deepthi et al. (2024) evaluates Apache Flink for real-time analytics and reports sustained stream-processing throughput of about 9.82 MB per second on commodity hardware. Alsalama et al. (2025) extends Apache Spark with approximate query processing and stratified sampling for large-scale geospatial joins.

The third stream is machine learning applied to communications. Gong et al. (2026) combine long short-term memory with temporal convolutional networks for per-beam LEO traffic prediction and report an 82 percent reduction in mean absolute error relative to classical baselines. Hwang et al. (2020) train an unsupervised 1D CNN autoencoder (D-PACK) that flags malicious flows after inspecting two packets and 80 bytes, at about 99 percent accuracy. Liu (2020) applies AdaBoost with back-propagation and k-means to a smaller satellite-telemetry dataset at 70.67 percent accuracy. Parvini et al. (2024) use multi-agent reinforcement learning with twin delayed deep deterministic policy gradients for vehicle-to-everything resource allocation, a problem structurally similar to beam and spectrum assignment.

The fourth stream is edge computing and 5G/6G integration. Mohammed et al. (2022) propose the SEC-5G architecture in which lightweight analytics run on-board satellites and at gateway edge nodes, reducing end-to-end application response time from about 52 seconds to 1.1 milliseconds for representative workloads.

3. Technical Details

The technical setting is defined by three characteristics of satellite telemetry: volume, velocity, and variety. A constellation of one thousand LEO satellites, each with sixteen active beams emitting per-beam counters once per second, generates on the order of tens of thousands of events per second from counters alone, before flow records or modem-level

signal-quality data are added. Ochuba et al. (2024) and Liu (2020) place the annual volume in the petabytes, and Gong et al. (2026) note diurnal and beam-level bursts that stress any fixed-capacity pipeline.

3.1 Orbital regimes and delay budgets

Geostationary satellites at about 36,000 kilometres altitude incur roughly 120 milliseconds of one-way propagation delay, while medium-Earth-orbit satellites between 5,000 and 25,000 kilometres and LEO satellites between 500 and 900 kilometres have one-way delays between 1.5 and 30 milliseconds (Kodheli et al., 2021). These values set the constraints on any control loop that closes over a ground-based analytics platform.

3.2 Platform roles

Apache Kafka serves as the ingest layer. Partitioned topics, broker-side replication, and producer batching allow it to absorb millions of telemetry events per second (Vyas et al., 2022). Apache Flink provides streaming analytics with event-time windowing and keyed state, suitable for per-beam sliding-window aggregates and online anomaly scoring (Deepthi et al., 2024). Apache Spark provides batch and geospatial analytics over historical telemetry, with geohash partitioning and approximate query processing keeping large ad-hoc queries tractable (Alsalama et al., 2025).

3.3 Machine-learning families

Three ML families appear in the literature. Supervised and sequence models (LSTM plus TCN in Gong et al., 2026; AdaBoost-BP in Liu, 2020) predict per-beam traffic and fault conditions. Unsupervised models (the 1D CNN autoencoder of Hwang et al., 2020) detect novel attack or fault patterns without labelled data. Reinforcement learning (the MARL-TD3 formulation of Parvini et al., 2024) handles sequential decisions such as beam and spectrum allocation. A reference architecture has to host all three.

4. Obstacles

Several obstacles stand in the way of a unified architecture. The first is on-board resource scarcity. Space-qualified platforms such as NVIDIA Jetson or Xilinx Zynq FPGA modules used by Mohammed et al. (2022) run under tight power and thermal budgets, which limits the size of any model placed on the satellite. The second is data heterogeneity. Telemetry formats, sampling intervals, and clock quality differ across vendors and generations of hardware, which complicates any schema that spans the whole constellation (Ochuba et al., 2024). The third is the non-IID nature of per-satellite data, which degrades simple federated averaging. Parvini et al. (2024) observe similar effects in vehicular networks, where local distributions differ enough to destabilise training. Finally, radiation-hardened hardware is behind commercial silicon by several generations, so any proposed on-board component has to be simple enough to run on older, slower chips.

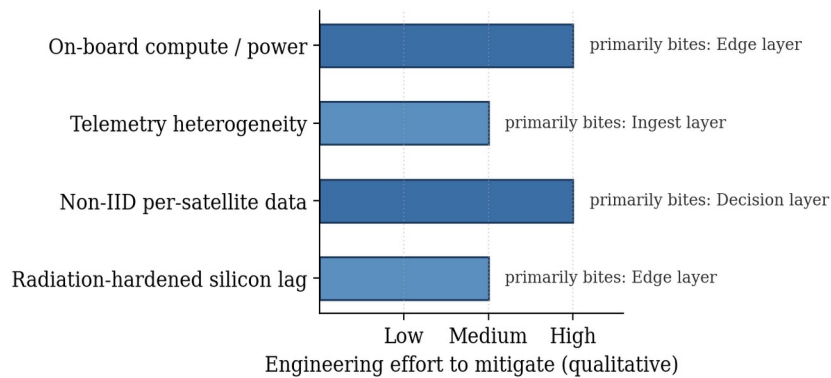


Figure 1. Four obstacle categories ranked by qualitative engineering effort to mitigate, with the architectural layer where each obstacle primarily manifests.

5. Promise

Despite these obstacles, results in the literature suggest the payoff is large. Gong et al. (2026) report an 82 percent reduction in mean absolute error for LEO per-beam traffic prediction, which translates into tighter capacity planning and less overprovisioning. Hwang et al. (2020) achieve about 99 percent accuracy on early-stage network anomaly detection after inspecting only two packets and 80 bytes, enabling detection at the edge before attacks

propagate. Mohammed et al. (2022) demonstrate a reduction in end-to-end application response time from about 52 seconds to 1.1 milliseconds by moving analytics to the network edge. Deepthi et al. (2024) sustain 9.82 megabytes per second of stream processing on commodity hardware with Flink, which is already within an order of magnitude of the per-satellite counter rates described in Section 3. Taken together, these numbers show that the individual pieces of a data-driven control loop are already mature.

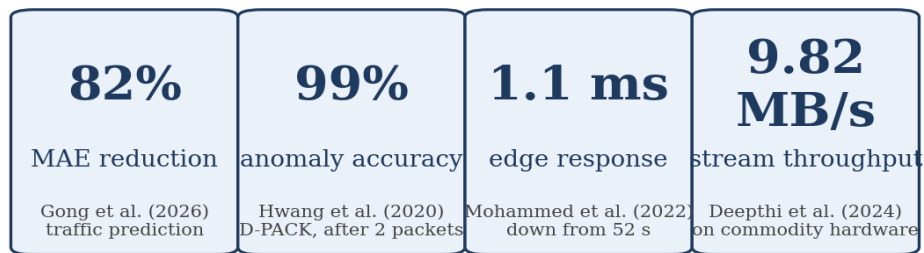


Figure 2. Headline quantitative results from the literature that this paper integrates: traffic-prediction error reduction, early-stage anomaly accuracy, edge response time, and sustained stream throughput.

6. Suggested Course of Action

The suggested course of action is to adopt a layered reference architecture and use the joined feature stream it produces. The architecture has four layers. An ingest layer built on Apache Kafka, a streaming-analytics layer built on Apache Flink, a batch and geospatial layer built on Apache Spark, and a decision layer in which machine-learning models are trained offline, pushed to streaming jobs, and selectively mirrored to on-board satellites following the SEC-5G pattern of Mohammed et al. (2022). Table 1 summarises the platform mapping. Figure 3 gives the corresponding stack diagram with the data-flow loop closed through the on-board edge.

Layer	Platform	Workload	Reference
Ingest	Apache Kafka	Telemetry, flow records	Vyas et al. (2022)

Layer	Platform	Workload	Reference
Streaming	Apache Flink	Windowed features, online scoring	Deepthi et al. (2024)
Batch / Geo	Apache Spark	Model training, AQP, geohash joins	Alsalama et al. (2025)
Edge	SEC-5G on Jetson / Zynq	On-board inference, local mitigation	Mohammed et al. (2022)

Table 1. Mapping of architecture layers to platforms and representative workloads.

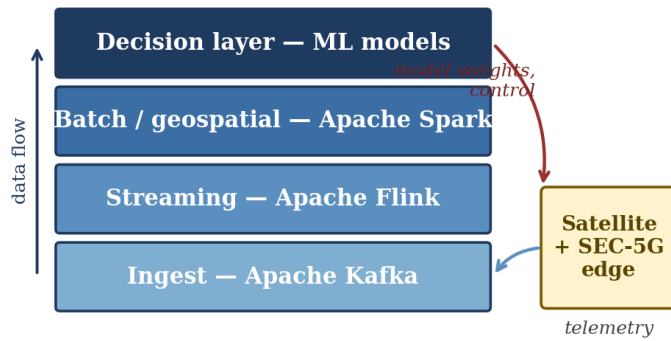


Figure 3. Layered reference architecture. Telemetry from the satellite enters at the Kafka ingest layer; Flink and Spark turn it into features and trained models; the decision layer pushes model weights and control messages back down to the SEC-5G edge on the satellite, closing the loop.

Two pipelines make the architecture more concrete. For per-beam DDoS detection across a thousand-satellite, sixteen-beam-per-satellite constellation, Kafka partitions on a (satellite, beam) key, Flink maintains sliding one-minute windows, and a D-PACK-style autoencoder (Hwang et al., 2020) scores each window. Anomalous beams trigger a control message that, where SEC-5G mirrors the model on-board, the satellite executes within milliseconds. For capacity planning, the same Kafka stream feeds a Flink and Spark pipeline that retrains the LSTM plus TCN predictor of Gong et al. (2026) weekly on geohash-partitioned history. Both pipelines run on commodity hardware. Both also show what is missing.

Together, the two pipelines and the four-layer stack of Figure 3 cover every component family identified in Section 2. Vyas et al. (2022) at the ingest layer, Deepthi et al. (2024) at the streaming layer, Alsalama et al. (2025) at the batch and geospatial layer, the supervised, unsupervised, and reinforcement-learning models of Gong et al. (2026), Hwang et al. (2020), Liu (2020), and Parvini et al. (2024) at the decision layer, and Mohammed et al. (2022) at the on-board edge. Adopting this architecture is therefore not an abstract proposal. It is an integration of results that the literature has already validated.

7. Conclusion

This paper has surveyed the four research streams relevant to AI-driven satellite-network optimization (survey work, big-data platform engineering, machine learning, and edge computing) and argued that integrating their components into a layered reference architecture is the most useful near-term direction. The expected operational and economic implications follow. Operationally, combining the 99 percent early-stage anomaly detection of Hwang et al. (2020) with the 1.1-millisecond edge response of Mohammed et al. (2022) would allow DDoS and rogue-beam events to be contained on-board within a single round trip rather than after seconds of damage. Economically, the 82 percent traffic-prediction MAE reduction of Gong et al. (2026) would translate into less capacity overprovisioning and savings in equivalent transponder cost at constellation scale. For end users, dropped packets at handover events, today's largest source of short-duration LEO outages, would fall without any change to user equipment. And because every layer of the proposed architecture is open-source and commodity, the engineering bar for new entrants would drop, allowing smaller operators to compete with mega-constellations on operational sophistication. The contribution of this paper is therefore not a new algorithm but a synthesis. It is an architectural account of how the existing components of the literature can be assembled into a working data-driven control loop.

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